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M 402 MAKE UP WATER SYSTEM

1 SCOPE OF WORK

The Bidder shall supply the following in accordance with part 4, Subsection 1.2. “Summary of Work”.

The make up water tanks complete with accessories, including but shall not be limited to the following major components.

- Tank
- Nozzles
- Manholes
- Airtight seal material (Inner floating type)
- Spiral stairway and handrail
- Valves, piping and flexible joint
- Pumps and Motors
- Foundation bolts and nuts
- Instrumentation and control equipment
- Electrical equipment
- Painting
- Cables
- Electrical equipment

2 CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4. Subsection 2.5 “Codes and Standards”.

The codes and standards extracted from the Part 4, Subsection 2.5 “Codes and Standards” are specifically applicable to the design manufacturer and testing of the equipment included in this specification.

AWWA	American Water Works Association
AISC	American Institute of Steel Construction
AISI	American Iron and Steel Institute
ANSI	American National Standard Institute
API	American Petroleum Institute
ASTM	American Society of Testing and Materials
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
HIS	Hydraulic Institute standard
NEMA	National Electrical Manufacturers Association
SSPC	Structural Steel Painting Council

3 DESIGN AND OPERATING CONDITIONS

One (1) new make up water tank shall be installed for block 3 & 4. New make up water tank will be connected to existing make up water tank. The capacity of tank shall be 2000 m³.

Six(6) make up water pumps (2 units for block 2, and 2 units for block 3, and 2 unit for block 4) shall be provided to serve the water supply from Make up Water Tanks. These pumps shall capable to fulfill the requirements of make up water for block 3 & 4. The inlet make up water pumps shall be connected to the suction header and the outlet make up water pumps shall be connected to discharge header completed with auxiliaries. The bidder shall supply a blind flange at suction header for future connection from block 1 and supply a blind flange at discharge header for future connection to block 1.

ADD2
-IIC
-001

Level control valve shall be installed at the inlet of tank to maintain the tank water level.

The make up water pump shall transfer the make up water from the make up water tank to surface condenser during operation of the plant and also for filling up the Steam Turbine during initial operation of the plant.

Each Make up water pump shall be provided with a continuous flow recirculation line connected to the make up water tank to provide the minimum flow through the pumps. Minimum recirculation flow shall be added to the system requirements to determine pump design flow.

Capacity of make up water pumps shall include the maximum make up water requirements for Steam Turbine on Block 3 and Block 4, regeneration water for Demineralization Plant, make up for closed cooling water system, and other miscellaneous users. These pumps shall be arranged for automatic start and stop.

Tank capacity calculations shall be submitted to the Owner for approval.

4 DETAILED REQUIREMENT

4.1 Tank

The tank shall be of all welded construction designed and fabricated in accordance with API 650 including appendix E, seismic zone 4, and shall be supplied complete with pipe connections and manholes.

The plate thickness of the tank shell and roof shall be not less than 6.5mm. Tank bottom course shall be 8 mm minimum. Plate material shall be ASTM A283 Grade C.

Bottom plates shall be of lap-welded construction.

Shell plates shall be of butt-welded construction.

Roof plates shall be of lap-welded construction, and shall be self-supporting spherical type.

The tank shall be provided with the following:

- Inlet nozzle
- Outlet nozzle
- Return nozzle
- Overflow nozzle
- Blow off nozzle
- Level transmitter nozzle
- Minimum flow nozzle

- Vacuum relief nozzle
- Two (2) Spare nozzles
- Sample nozzle

The sufficient number of manholes shall be installed on the roof plate and shell plate.

The spiral stairway to the tank roof shall be installed.

The handrail and platform shall be installed on the roof.

The grounding wiring shall be installed.

The tank atmospheric vent shall be fitted with an air filter of replaceable type to filter out atmospheric dust down to one (1) micron in size with a suitable area to maintain an air velocity below 150 m/sec at maximum tank filling or draining rates.

Anchor bolts shall be provided.

4.2 Pump

Six (6) Capacity Horizontal centrifugal type pumps (**2 units for block 2, 2 unit for block 3 and 2 units for block 4**) shall be installed in accordance with Part 4, Technical Specification M-006. ADD2
-IIC
-004

Pumps shall be installed out door at the pump station near the Make-up Water Tanks.

The motor shall be in conformity with Electrical Technical Specification E-210.

4.3 Piping, Valves and Flexible Joint

Anchor bolts shall be provided.

Valves shall be provided at the following nozzles:

- Inlet nozzle
- Outlet nozzle
- Return nozzle
- Blow off nozzle
- Level transmitter nozzle

- Minimum flow nozzle
- Two (2) Spare nozzles (with blind flange)
- Sample nozzle

The valves shall be stainless steel 10 Bar rating, flanged type.

The overflow piping shall be led to trench.

Flexible joint of stainless shall be installed on inlet and outlet connection.

Stainless steel make up supply piping shall be provided from the make up water tank to combine cycle plant.

Level control valve with stop valves shall be installed on the inlet piping of tank, and shall be in accordance with Part 4 Technical Specifications I- 001 “Instrumentation and Control”.

The piping shall be in conformity with Part 4, Technical Specifications M-001 “Piping”.

The Valves shall be in accordance with Part 4, Technical Specifications M-004 and M-005.

4.4 Painting

Both inner and outer surfaces of all tank plates shall be shot-blasted to remove mill scale, and shall be painted by one (1) layer of zinc-enriched primer before shipment.

After the installation of tank, welded areas and rusted parts of shop painted plates shall be re- painted by primer paint after the removal of rust scale by a grinder.

After the above work, epoxy resin base with 300 micron in thickness shall be painted on the inner surface and phenolated alkyd resin, base paint with 150 micron in thickness shall be painted on the external surface.

Painting shall be per Part 4, Specification M-001, S-420 and S-430.

5 PERFORMANCE TESTING AND GUARANTEE

The Bidder shall conduct performance testing in accordance with Part 4, Subsection 2.6 Testing and Guarantees and shall guarantee the following system guarantees:

As minimum the Bidder shall perform all the test as required but not be limited to the following:

- Leak testing
- Radiography
- Liquid penetrant examination
- Magnetic particle examination

The Bidder shall execute the following shop tests and shall submit the test reports to the Owner.

- (a) Material
- (b) Dimension check

The Bidder shall execute the following field tests and shall submit the test reports to the Owner.

- (a) Dimension check
- (b) Water filling test (hydrostatic test)
- (c) Penetrant and X-Ray test
- (d) Settlement check of tank base
- (e) Inspection for painting

6 QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality assurance/quality control shall conform to the requirements of Part 4, Subsection 2.7.

The quality assurance/quality control levels for the respective tank shall correspond to the levels imposed on the equipment or system terms of which the respective tanks are a part thereof.

The quality assurance/quality control procedures so specified in Part 4, Subsection 2.7 and shall include, but not be limited to the following:

- Welding
- Nondestructive Testing
- Heat treating
- Painting
- Packaging and shipping

The quality control documentation as specified in Part 4, Subsection 2.7 shall be as required by the applicable codes.

Unless required by the applicable ANSI, ASME or API code, ultrasonic magnetic particle or liquid penetrant examinations are not required.

6.1 Radiography

All radiography shall be in accordance with the procedures and comply to the acceptance standards referenced in API 650.

Spot radiography shall be modified as follows :

- a. One radiograph shall be taken in the first 3 m of weld and one spot shall be examined for each additional 15 m of full-penetration butt welds, or fraction thereof.
- b. At least 2 intersections of vertical and horizontal joints, when they exist, shall be included in the radiographs.
- c. All films shall be subject to the Owner's review and comment.

6.2 Leak Testing

The tank shall be leak-tested following all fabrication, but prior to painting, coating or lining the interior.

The tank shall be filled with water and inspected frequently during filling.

If water is not available for testing, all joints shall be tested with a vacuum box and soap solutions. The gauge on the box shall register the vacuum and the differential pressure across the joints shall be at least 0.35 Bar.

The bottom joints of flat-bottomed tanks shall be tested with a vacuum box and soap solution prior to hydro testing.

Any leaks detected during hydro testing or vacuum testing shall be repaired.

Bidder shall provide (and remove after use) necessary test pump and temporary piping, valves, blind flanges, etc, for filling and draining the tanks.

Result of all test shall be documented and submitted to the engineer with a sketch showing areas examined.

Test procedures shall be submitted for review to the owner. No testing work shall be done until these procedures are acceptable to the owner.

7 INSTRUMENTATION AND CONTROL

The instruments and device shall be provided by the Bidder but not limited to the following:

Level transmitters and remote indicator for indication on the panel in demineralized water treatment plant control panel and central control room.

Dial type level indicator with high and low level switch for annunciator on main control panel.

Level switches for water treatment equipment interlock and make up water pump interlock.

Refer Part 4, Technical Specification I-001 for the requirements for the Instrument and Control systems devices and I-002 for the requirement for the Local Control Panels.

8 ELECTRICAL SPECIFICATIONS

8.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for the Make-up Water System.

Those equipment shall include, but not be limited to, the followings :

- (a) Motors
- (b) Power panel
- (c) Conduits, cables and wiring.

- (d) All electrical control, monitoring and instrumentation devices, and other appurtenances required for this specification, shall be provided by the Bidder.

For general electrical requirements, see Part 4, Technical Specification E-001.

8.2 Motors

For detailed electrical requirements, see Part 4, Technical Specification E-210 for Low Voltage Motors.

8.3 Power Panels

For detailed electrical requirements, see Part 4, Technical Specification E-001 and E-280.

The panel shall have a degree of protection IPW 55 of IEC 60529.

8.4 Conduits, Cables and Wiring

All motors and electrical, control, monitoring and instrumentation devices furnished and installed by the Bidder shall be wired by him to the power and local control panel respectively.

The wiring between those panels shall also be included.

The required power, control, instrumentation and thermocouple extension cables shall be provided by the Bidder.

For detailed electrical requirements, see Part 4, Technical Specifications E-230, E-231 and E-232.

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